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SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Geoengineering, programmable medicines, COVID-19 updates, our ignorance about saliva, how galaxies evolve and the latest research in the field of microbots...

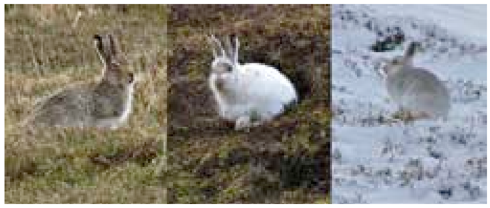
Probing the universe

Researchers have developed a new framework to use multi messenger astronomy to probe the most extreme events in the universe, such as two colliding neutron stars. The energies and physics in these interactions can reveal much about the fundamental nature of the cosmos.
http://dgit.in/colstar

WHAT'S NEW

Climate change is exposing bunny rabbits to predators

As the winters get less colder with every passing year in Scotland, the snow in the mountains reman for a shorter duration. The rabbits here have a summer coat and a winter coat, which is an adaptation for camouflage to make the rabbits harder to see by predators. However, because the snow is melting too fast, rabbits with bright white coats are sticking out like a sore thumb. The research is distressing for conservationists and environmentalists as it indicates that there are species that shed colours that are not able to evolve rapidly enough to keep up with the changes in seasons brought about by human industrial activity. The evolutionary arms race between prey and predator is balanced on the hair of the hare, and though it's not a crisis for the rabbits just yet, it could soon turn out to be!
http://dgit.in/gwbny



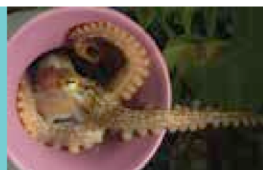
Voyager spacecraft continue to advance science

The 40-year old planet hopping journey through the Solar System that allowed scientists to first probe distant bodies beyond Mars, continues to advance scientific understanding on its journey through interstellar space. Voyager still periodically beams back observations from the interstellar medium, where no spacecraft has reached before. Both Voyager 1 and 2 spacecrafts have detected high energy

electrons that have been accelerated by coronal mass ejections. These electrons are accelerated to nearly the speed of light, about 670 times faster than the shockwaves that initially pushed them, and move along interstellar magnetic field lines. The Voyager spacecraft first detect the high energy electrons, which is then followed by much slower moving electrons days later, and finally, particles from the shockwave itself are detected at times, as long as a month after the initial detection. The Voyagers are the only instruments capable of studying this.
http://dgit.in/voysci

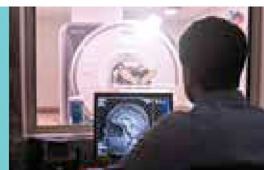
Deep space brine

Modelling of the interior of Europa has shown that some of the cryovolcanic eruptions can originate from deposits of brine accumulated close to the surface, which would make the matter from the spouts ideal for investigating for signs of life, by future interplanetary missions.
http://dgit.in/eubrine



Octopuses can flex

It's official, scientists have declared that the arms of the octopus are the most flexible limbs known to nature. Researchers recorded 16,563 examples of unique arm movements, which is a combination of how the animals bend, twist, elongate and shorten their arms.
http://dgit.in/octarm



Neuroscience of creativity

What goes on in our heads when we are inspired and write a screenplay or start a new business venture? Researchers investigate the neurological activity linked to creativity by studying musicians while they're in the process of improvisation.
http://dgit.in/brainbulb